

Control Statements



traffic signals for your code



which part of your

code should run &

when

① Conditional Statements ✓

↳ if, elif, else

② Looping statements ✓

↳ for, while

③ Control transfer statements ✓

↳ break, continue, pass

i = 1

```
while i <= 10:
```

```
    if i == 5: true
```

```
        break
```

```
    print(i)
```

```
    i = i + 1
```

↳ out of the while loop

i = 7

↳ break

i = 1

1 ✓

i = 2

2 ✓

i = 3

3 ✓

i = 4

4 ✓

i = 5

```
for i in range(11):
```

```
    if i == 7:
```

```
        break
```

```
    print(i)
```

i = 0

0

i = 1

1

i = 2

2

i = 3

3

i = 4

4

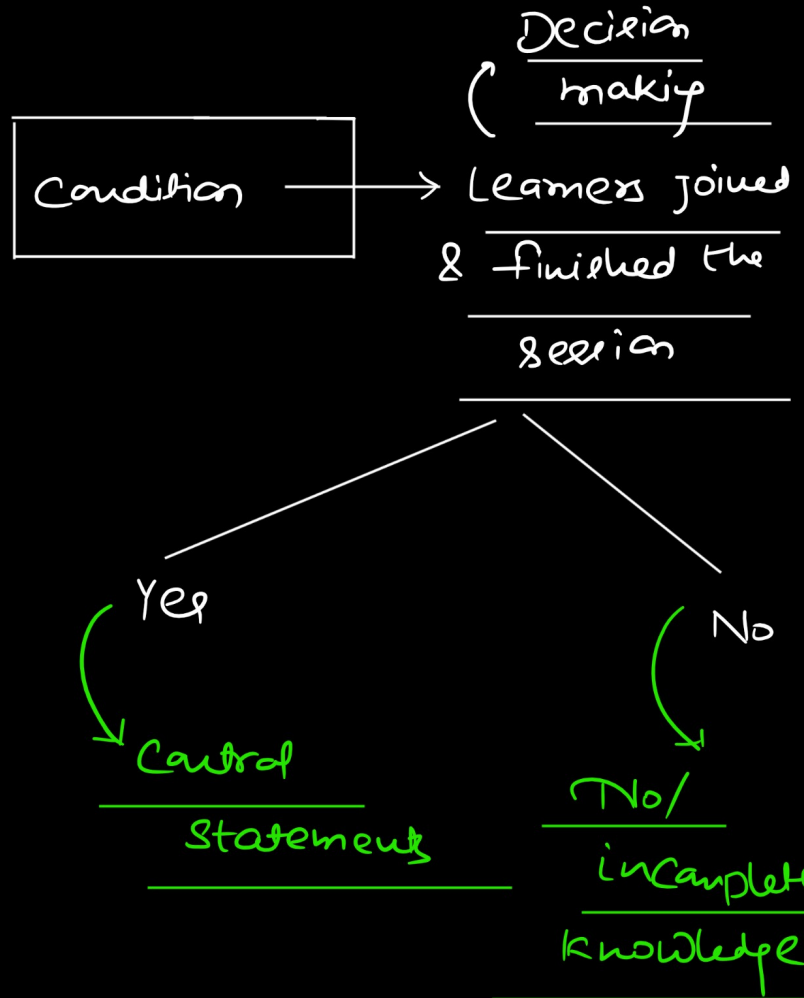
i = 5

5

i = 6

6

Conditional Statements



↳ if

↳ if - else

↳ if - elif - else

x = -10

if x > 0: print('x is positive')

else: print('x is negative')

True / false

Time complexity → how much time the code is taking to execute.

Conditional Statements → O(1) Constant time Complexity

```
age = int(input("Enter your age: "))
```

```
has_id = True
```

```
if age >= 18:
```

```
    if has_id:
```

```
        print('You can enter the club.')
```

```
    else:
```

```
        print('You need an ID to enter.')
```

```
else:
```

```
    print('You are too young to enter')
```

internal

ch → '1' → Contains

0	1	2	3	4	5	6	7	8	digit
P	y	t	h	o	n	1	2	3	or
↑	↑	↑	↑	↑	↑	↑			not

Note:

Statement	Description
if	Executes code when condition is true
if-else	Executes one block if true, another if false
if-elif-else	Handles multiple conditions
nested if	Condition inside another condition

```
password = input("Enter your password: ") ✓
```

```
password_length = len(password) ✓
```

```
has_digit = any(char.isdigit() for char in password)
```

```
if password_length >= 8 and has_digit:
```

```
    print("Strong password")
```

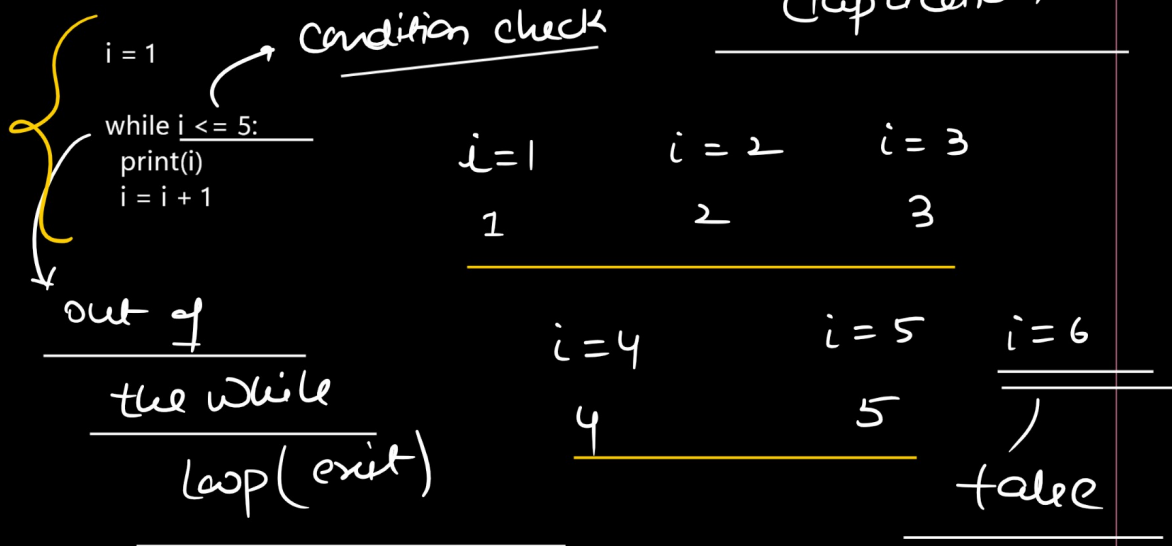
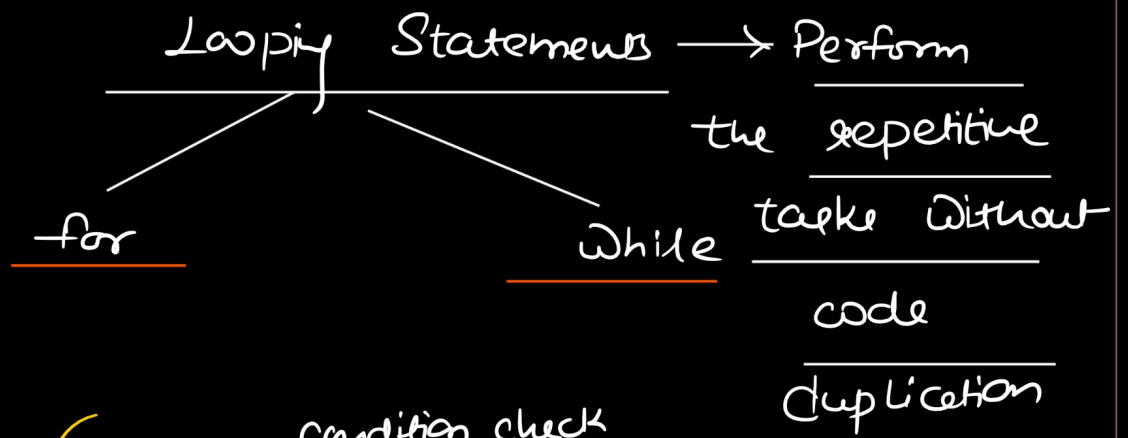
```
elif password_length >= 8 and not has_digit:
```

```
    print("Weak password: Add a digit in your password")
```

```
else:
```

```
    print("Weak and too short password")
```

check whether password
contains any digit
or not



Break → stopping the entire loop early

Continue → skip the current iteration

Scenario	Ideal Control Statement	
Stop looping early	break	✓
Skip unwanted iterations	continue	✓
Placeholder for incomplete logic	pass	✓

don't know the logic
presently

Nested

loops

```
for i in range(1, 6):  
    for j in range(1, 11):  
        print(i * j)
```

outer loop

inner loop

print the

table of

1 to 5

$i = 1$

$j = 1$

$j = 2$

$j = 10$

(1, 2, 3, 4, 5, ..., 10)

$i = 2$

$j = 1, j = 2, \dots, j = 10$

(2, 4, 6, 8, ..., 20)

$i = 3$

$j = 1 \dots j = 10$

$i = 4$

$j = 1 \dots j = 10$

$i = 5$

$j = 1 \dots j = 10$